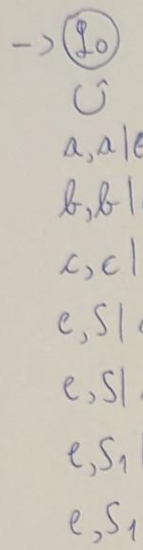
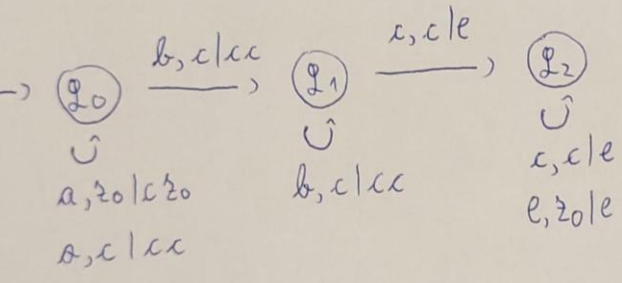
1. a. $S \rightarrow aSc \mid aS_1c$

$S_1 \rightarrow bS_1c \mid bc$

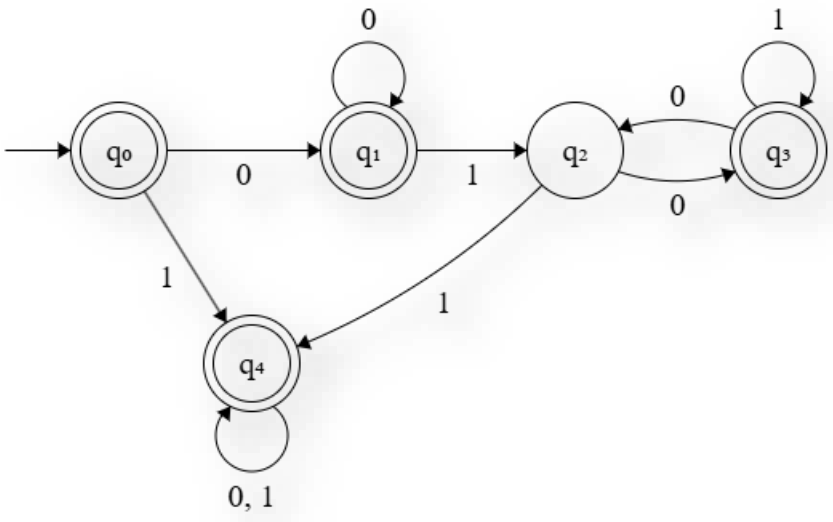
b.



c.



2.a



2.b. $S \to 0S_1 \mid 1S_4 \mid e \Rightarrow S = 0S_1 + 1S_4 + e$

$S_1 \to 0S_1 \mid 1S_2 \mid e \Rightarrow S_1 = 0S_1 + 1S_2 + e$

$S_2 \to 0S_3 \mid 1S_4 \Rightarrow S_2 = 0S_3 + 1S_4$

$S_3 \to 1S_3 \mid 0S_2 \mid e \Rightarrow S_3 = 1S_3 + 0S_2 + e$

$S_4 \to 0S_4 \mid 1S_4 \mid e \Rightarrow$ ~~$S_4 = (0+1)^*$~~ $S_4 = 0S_4 + 1S_4 + e \Rightarrow S_4 = (0+1)^*$

$S_3 = 1S_3 + 0(0S_3 + 1S_4) + e$

$S_3 = (1 + 00)S_3 + 01S_4 + e$

$S_3 = (1+00)^* [01(0+1)^* + e]$

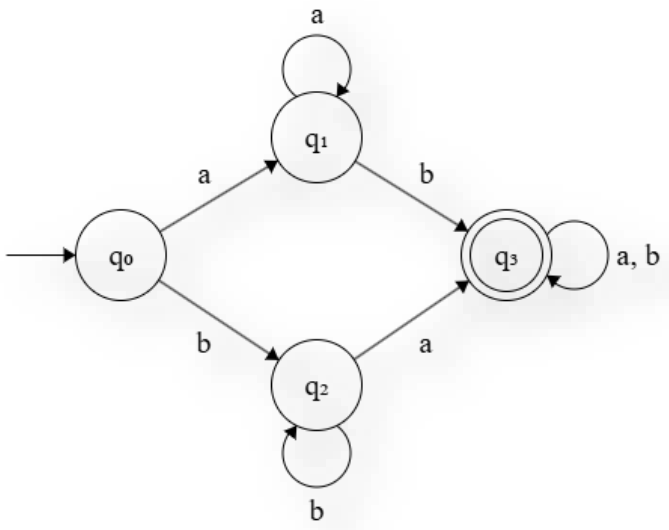
$S_2 = 0(1+00)^* [01(0+1)^* + e] + 1(0+1)^*$

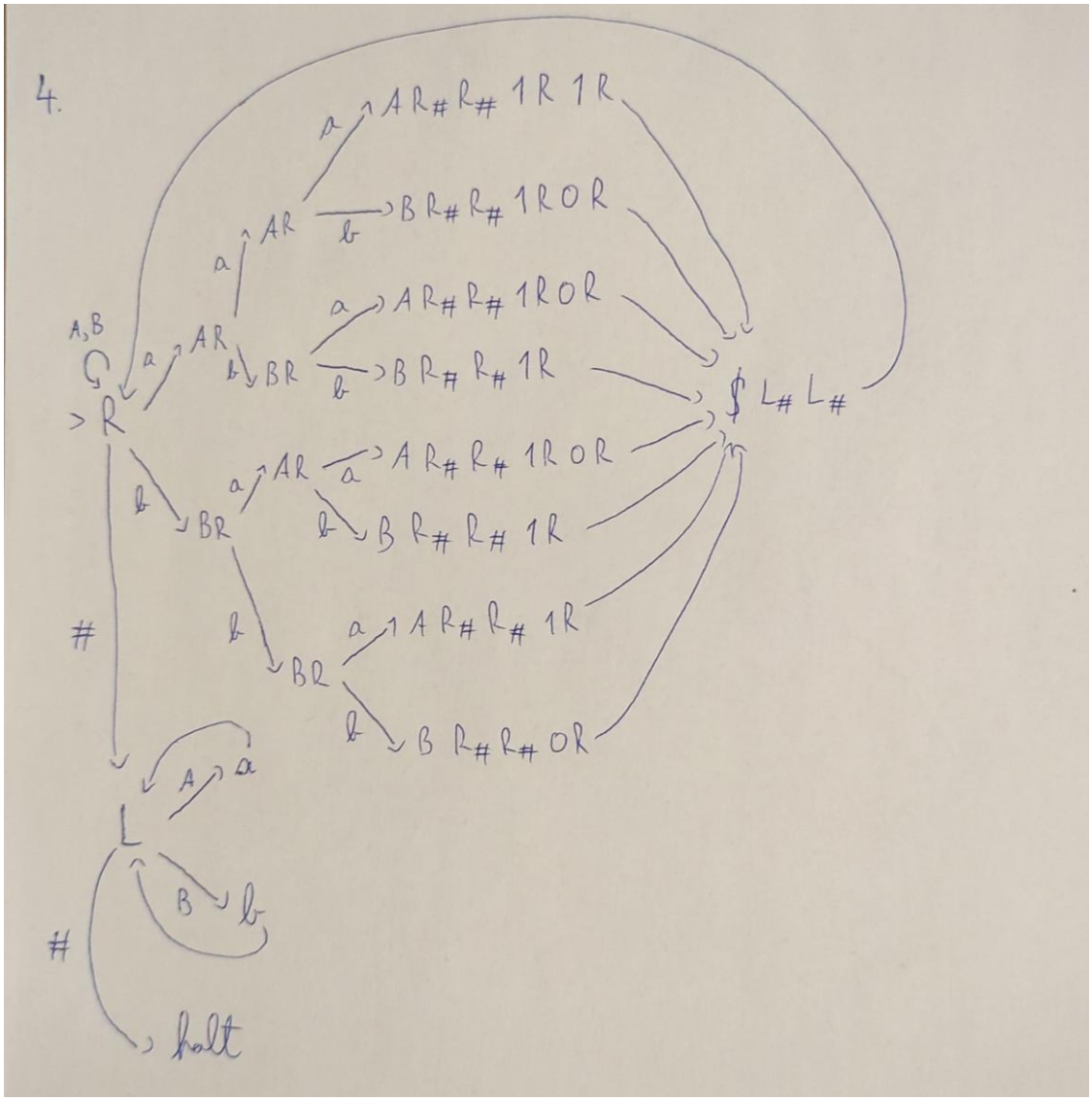
$S_1 = 0S_1 + 1S_2 + e \Rightarrow$

$S_1 = 0^* \{10(1+00)^* [01(0+1)^* + e] + 11(0+1)^* + e\}$

$S = 0^+ \{10(1+00)^* [01(0+1)^* + e] + 11(0+1)^* + e\} + 1(0+1)^* + e$

3




4.
A,B
> R
a
b
AR
BR
a AR # R # 1R 1R
b BR # R # 1R 0R
a AR # R # 1R 0R
b B R # R # 1R
a AR
a A R # R # 1R 0R
b B R # R # 1R
b
BR
a A R # R # 1R
b B R # R # 0R
#
A a
L
B b
#
halt
$ L # L #